

Sparse Transformer with Locality-Biased Attention for Document Classification

Anon

2024

DOI (synthetic): 10.9999/trap.2024.004

Abstract

We propose a locality-biased sparse Transformer for long-document classification. We benchmark against five strong baselines (Longformer, BigBird, Linformer, Performer, vanilla Transformer) on three datasets. Our method achieves new SOTA on HyperNews-10K while reducing memory by 38%.

1 Method

We replace full self-attention with a locality-biased kernel:

$$\text{Attention}_{\text{ours}}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + \lambda B\right) V \quad (1)$$

where $B_{ij} = -|i - j|/s$ is a learned locality prior with scale parameter s , and λ balances the standard attention against the locality bias.

2 Results

Table 1: F1 scores on three benchmarks (higher is better). Bold = best in column.

Model	HyperNews-10K	LegalDoc-5K	BioMed-20K
Vanilla Transformer	72.1	68.4	64.2
Longformer	78.3	73.2	70.8
BigBird	79.0	74.1	71.5
Linformer	74.8	70.6	67.3
Performer	76.2	71.9	69.1
Ours (full)	81.4	74.8	71.9
Ours (no locality bias, $\lambda = 0$)	75.9	71.8	68.7
Ours (no sparse kernel)	79.2	73.5	70.9

Our full model achieves 81.4 F1 on HyperNews-10K, outperforming all baselines. Ablations show that removing the locality bias ($\lambda = 0$) drops F1 by 5.5 points, and removing sparsity drops it by 2.2 points.

3 Discussion

Our method achieves SOTA with 38% less memory than BigBird.